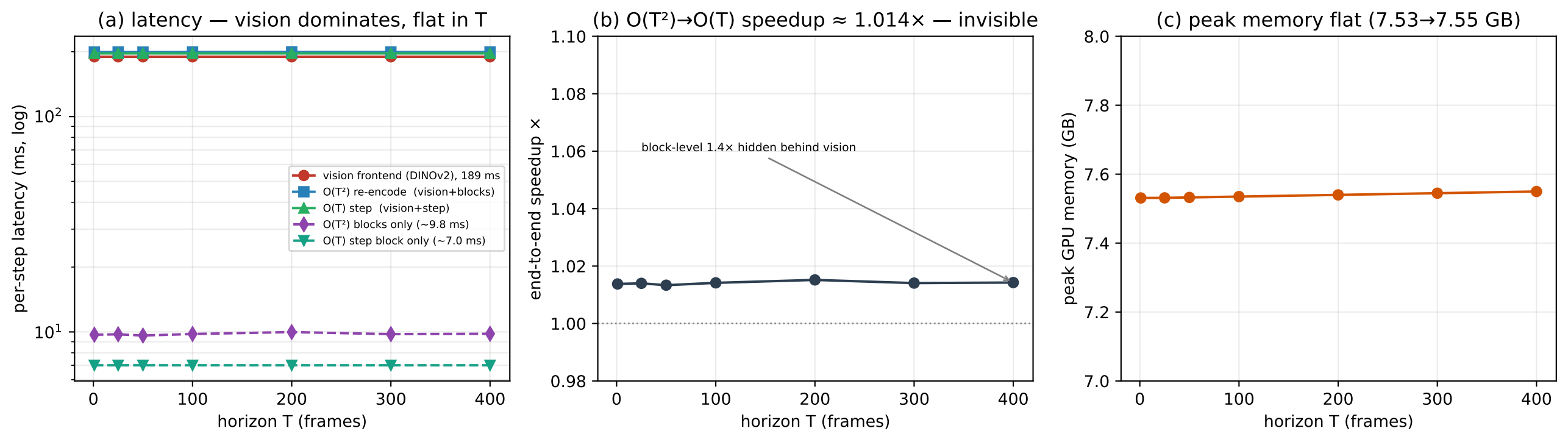
(a) latency — vision dominates, flat in T
per-step latency (ms, log)
10^2
10^1
vision frontend (DINOv2), 189 ms
O(T²) re-encode (vision+blocks)
O(T) step (vision+step)
O(T²) blocks only (~9.8 ms)
O(T) step block only (~7.0 ms)
0
100
200
300
400
horizon T (frames)
(b) O(T²)→O(T) speedup ≈ 1.014× — invisible
end-to-end speedup ×
1.10
1.08
1.06
1.04
1.02
1.00
0.98
block-level 1.4× hidden behind vision
0
100
200
300
400
horizon T (frames)
(c) peak memory flat (7.53→7.55 GB)
peak GPU memory (GB)
8.0
7.8
7.6
7.4
7.2
7.0
0
100
200
300
400
horizon T (frames)